A Comprehensive Literature Review on Landslide Detection in Remote Sensing

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Abstract

There are several sources of remote sensing (RS) imagery that are known for their ability to detect landslides. We review 40 articles published in the past two years that employed machine learning, OBIA, CNN and transformers and SAM-based models to detect or segment landslides. Multimodal data fusion (optical + DEM), attention-enhanced and hybrid CNN–Transformer architectures and advanced learning strategies (e.g., transfer learning, weak supervision and knowledge distillation) are among the key trends that have driven SOTA progress. R41 importantly, models that utilize local texture features as well as global contextual awareness (i.e., which pixel belongs to which area) have made segmentation more precise, especially when it comes to boundary and cross-region performance (R15), have made segmentation more precise. The review points to the need for standardized large-scale data sets and model interpretability for real-world operational use.

Keywords: Landslide Detection, Remote Sensing Imagery, Transformer Models, Deep Learning Segmentation

1 Introduction

As a major natural hazard, landslides take a large toll in human life and in economic loss. Traditional remote sensing (RS) imagery platforms that support large-scale monitoring rely on machine learning (ML) and/or object-based image analysis (OBIA) methods, which in turn rely on hand-crafted features to detect objects of interest. Recent studies have shown the potential of using hybrid CNN-Transformer networks and SAM-based AI models for landslide detection. However, these methods fail to integrate local (e.g., regional tectonic and lithologic attributes) and global features (e.g., climatic factors) to capture contextual relationships into their learning process. Limited standardized datasets, the so-called "black-box" interpretability problem and difficulty in operational deployment are likely to remain obstacles. In this review, we summarize 40 D.L.-powered (primarily C.N.N.- based) landslide detection studies from R.S. images published in the past five years.

2 Literature Survey

2.1 CNN Based Models

The authors perform a systematic comparison of two popular DL architectures, namely, Mask R-CNN and U-Net, in terms of their performance and generalization for automatic landslide boundary delineation in optical remote sensing images taken in di erent regions of the world. They adopted a two-stage approach, first training the models on the larger public LRSTTC data set and then fine-tuning them on the smaller QTP (Qinghai-Tibet Plateau) data set. To address overlaps in mask predictions for each landslide object, we add a post-processing module to Mask R-CNN. We find that Mask-RCNN is the best-performing model, with a less pronounced decrease in performance compared to any of the other models as the size of the training set is reduced. Both models performed well overall, but neither was able to accurately segment complicated features like rockfalls and ancient landslides — features that also occurred infrequently in the original data set. To improve performance, the authors write that a next step is to augment the training data sets with more examples of these scenarios, "providing CNNs the opportunity to learn an appropriate representation.". Our result will inform more large-scale landslide inventory mapping based on remote sensing imagery and highlight current limitations on landslide mapping.

Ullo et al. develop a novel approach to detect landslides in UAV digital images using the Mask R-CNN deep neural network and transfer learning. Our pipeline comprises three stages: the dataset preparation and image augmentation (Set A: 160 images, Set B: 121 images); transfer learning from COCO-pretrained weights to fine-tune the model on our small dataset; and the use of the ResNet-50 and ResNet-101 as backbone networks. As a result, the full training procedure of the hybrid Mask R-CNN framework does not require a large number of images for the model to be able to recognize landslide characteristics from images. On the 160 images of Data Set A, the model using a ResNet-101 backbone exhibited its best performance in terms of Landslide-class metrics, achieving a Precision = 1.00, Recall = 0.93 and F1 measure = 0.97. The main limitation of their method is the requirement of training images with manually

(and therefore, labor intensively) built labels, the authors note. Instead of an object detection scheme, this problem requires an object classification approach.

Another study by Qin et al. explores the use of an improved Faster R-CNN deep learning model to quickly detect hazards in landslides. By replacing the VGG16 and ResNet50 feature extraction network with the new DarkNet53. Because there are a limited number of landslide samples in the data set, we also incorporated a data augmentation approach. To this end, we propose a novel remote sensing image object detection network called the R-AFNet. We illustrate the state-of-the-art efficacy of our algorithm with a multitude of experiments on a farmland pest dataset. The main limitations ultimately arise from the vanishing gradient problem that occurs with deeper networks, and the limited number of training samples available for more accurate models.

This study by Meena et al.[1] investigates the potential of U-Net and conventional machine learning (ML) models (SVM, KNN, RF) for automated landslide detection in the Nepali Himalayas using a remarkably small dataset (239 samples). The core finding was that the U-Net architecture slightly outperformed ML models on this restricted dataset due to its ability to preserve better structural connectivity and generate smoother results despite limited training samples. According to Meena et al. Adding altitude and slope (ALOS-PALSAR) data to the optical imagery improved the separation of the landslides from settlements and sand bars but did not significantly increase classification accuracy. The research confirms that despite the high potential of U-net, especially about result smoothness, deep learning and machine learning models remain highly comparable when severely limited sample sizes are utilized.

The paper authored by Cai et al.[2] aims to address the general dilemma of insufficient application, scarce samples and poor efficiency of deep learning-based landslide recognition method. The technique entailed superimposing 12 critical factors, including geological and topographic, onto the original satellite images to enrich the feature space. The analysis also shows that the integration of environmental factors significantly boosts model accuracy. About metrics, the optimal modified DenseNet model achieved the highest kappa at 0.9474 and F1 score of 0.9505 compared to the traditional ML and CNN models. The primary limitations/research gaps that motivated the study were the common challenges faced in deep learning method application, sample scarcity and inefficiency in landslide recognition.

The early and foundational phase of deep learning approaches for landslide detection, utilizing CNNs for pixel-level classification and instance segmentation. Core contributions involve applying established models, such as U-Net and its variants, sometimes in conjunction with traditional machine learning techniques (e.g., SVM or Random Forest) to show basic segmentation capabilities when sample sizes are small.[4] Mask R-CNN [1, 2] enabled the implementation of high-precision transfer learning techniques for the instance segmentation task. The improved Faster R-CNN model's backbone (DarkNet53) for CNN core implementation focused on stability and speed, solving the vanishing gradient problem in ultra-deep networks.[3] However, the CNN's embedded nature prevents it from obtaining global information when it comes to feature extraction. And forces the research community to look for complementary

non-convolutional architectures because of the tendency to confuse landslides with geologically similar backgrounds.

The paper presents MRIFE, a novel semantic segmentation network designed specifically for the detection of relic landslides in high-resolution remote sensing images[3]. The authors identify key challenges in this domain, such as visual blur and the limited availability of labeled datasets, which hinder effective landslide detection. To address these issues, MRIFE employs a dual-branch architecture that integrates a mask-recovering module and an interactive feature enhancement branch. This architecture is further enhanced by self-distillation techniques and contrastive learning, which improve the model's ability to distinguish between landslide features and background noise. The feature enhancement branch of MRIFE includes a masking module that processes input images and a teacher-student network that performs masked feature modeling and semantic feature contrast enhancement tasks. This approach allows the model to effectively capture both small-scale detailed textures and large-scale morphological information by utilizing dilated convolutions with varying receptive fields. The authors also implement a supervised contrastive loss function to enhance the model's ability to differentiate between landslide edge features and non-landslide categories, thereby accelerating network convergence. The dataset used for training, validation, and testing is meticulously divided into a 6:2:2 ratio, with a total of 1080 samples for landslides and 360 for non-landslides in the training set. The performance of MRIFE is evaluated using pixel-level semantic segmentation metrics, including pixel accuracy, precision, recall, F1-score, and mean intersection over union (mIoU). The results demonstrate significant improvements in performance metrics, with precision increasing from 0.4226 to 0.5347, mean IoU from 0.6405 to 0.6680, and F1-score from 0.5054 to 0.5646, showcasing the effectiveness of the proposed model in enhancing landslide detection capabilities.

The experimental results demonstrate that LDNet achieves a precision of 93.37 percentage recall of 91.93 percentage, and overall accuracy of 95.28 percentage, outperforming traditional models such as UNet and DeepLabv3+[4]. The study also includes an analysis of the effects of batch size on model performance, revealing that variations in batch size have a negligible impact on the results, although the GPU memory limitations necessitated a batch size of 2 for the experiments. Furthermore, the research emphasizes the importance of the attention mechanism in improving the accuracy and recall rates of the model, showcasing the effectiveness of CBAM in extracting deep features. The model's performance was validated across diverse geographical regions, indicating its potential for widespread application in landslide monitoring and disaster management. Overall, this study contributes to the advancement of semantic segmentation techniques in remote sensing and highlights the importance of optimizing model architectures for specific applications.

The study focuses on developing a Convolutional Neural Network (CNN) framework aimed at the semi-automatic detection of relict landslides in rainforest regions, which are often challenging due to dense vegetation[5]. To address the scarcity of labeled training data, the researchers employed data augmentation techniques, including horizontal and vertical flips, specifically targeting the positive class of landslide

data. The labeled dataset consisted of 390 tiles, with a split of 60 percentage for training (241 tiles) and 40 percentage for testing (149 tiles). The augmentation was crucial in balancing the dataset, as only the positive class was augmented to prevent skewing the results. The study also utilized k-means clustering to generate a comprehensive dataset, which was essential for the pre-training phase of the CNN, ultimately leading to improved accuracy in landslide detection. The results indicated that the proposed framework achieved recall rates exceeding 75 percentage across various combinations, although precision remained relatively low, ranging from 10 percentage to 20 percentage. This discrepancy highlights the inherent difficulties in detecting landslides within densely vegetated areas, where visual cues are often obscured. The findings suggest that while the CNN framework outperformed traditional methods in terms of accuracy, there is a pressing need for updated inventory maps and enhanced input data to further improve detection capabilities. Overall, the study underscores the potential of deep learning architectures in environmental monitoring and the importance of robust datasets for training effective models.

The study focuses on developing a hybrid CNN-transformer model, referred to as SCDUNet++, aimed at enhancing the accuracy of landslide mapping using remote sensing images[6]. The methodology involves deep transfer learning techniques to improve model performance, particularly in data-scarce regions. The results indicate that SCDUNet++ significantly outperforms existing models, achieving notable improvements in Intersection over Union (IoU) and F1 scores across various test areas. The findings suggest that the integration of spectral and topographic features plays a crucial role in enhancing mapping accuracy, while deep transfer learning effectively boosts the model's adaptability to new geographical regions. Overall, SCDUNet++ emerges as a practical and efficient solution for rapid landslide mapping in the aftermath of disasters, demonstrating its potential for real-world applications in disaster management and environmental monitoring.

2.2 Vision Transformer (ViT) Based Models

Zhao et al. [7] proposed a model, namely SwinLS to address the issues of landslide spatial correlation and class imbalance in remote sensing images. The authors adopted a two-stage training strategy and spectral selection process for the Swin Transformer model. The researchers determined the most beneficial spectral bands for the self-attention module through experimentation. They found RGB bands to be the optimal choice since multispectral inputs introduced noise and degraded performance. To address the imbalance between landslide and non-landslide pixels, they used a two-stage training method combining weighted cross-entropy and Lovász losses. This strategy aimed to make the model focus more on the landslide areas. The model was evaluated based on F1-score.The results show that SwinLS achieved an F1-score of 73.99 percentage. Additionally, they noted that data augmentation techniques (such as rotation and shifting) increased the F1-score by 4.2 percentage for the Swin Transformer model. The study's limitations and research gap relate to the loss functions' hyperparameter sensitivity (e.g., Focal loss), which would degrade model performance if the parameters were tuned incorrectly. The study's novelty lies in systematically combining spectral selection and a two-stage balanced training strategy to refine Swin

Transformer for landslide detection, offering preliminary insights into applying vision transformers to anomaly detection.

Huang and Chen [8] sought to solve the problem of long model runtimes and inefficiency by proposing a novel deep learning network structure for landslide recognition based on knowledge distillation. To this end, the authors created a new landslide sample database for Zigui County, Hubei Province and combined remote sensing images (RSIs) with nine landslide influencing factors (LIFs) such as elevation, slope and lithology. They developed what they termed a Distilled Swin-Transformer (DST) model, which leverages a larger and better-performing "teacher" AI model to train a smaller and more efficient "student" model. This method allows decreasing simulation costs without loss of accuracy. They compared various models, including ResNet50, Swin-Transformer and DeiT. Model efficiency was analyzed through the parameters like FLOPs and run time. The results demonstrated that the combination of RSIs and LIFs significantly increased OA, Precision, Recall, F1-score (F1) and Kappa. The DST model exhibited the best performance with an OA of 98.17 percentage, precision of 98 17 percentage, Recall of 98.17 percentage, F1 score of 98.16 percentage and a Kappa of 0.9766. An innovative aspect of the DST model is its efficiency, as it exhibited the lowest number of FLOPs (2.83 GFLOPs). The main limitation of this study is that the model's generalization ability was only tested on rainfall-induced landslides. It remains unknown whether it can perform equally well on other types, such as earthquake-induced landslides. The authors also mentioned that there is still room to further improve the model's efficiency in the pre-training stage.

ViT was the next obvious step because CNNs cannot model long-range dependencies due to their inherent design. The research primarily focuses on better harnessing the power of the Transformer model for remote sensing applications. The realization of this possibility during the development of SwinLS, [6] which achieved high accuracy by adapting the hierarchical structure of Swin Transformer to the task. The key debate arose here which data to use: SwinLS showed that sometimes it is better to utilize only RGB bands, because excessive multi-spectral noise dissipates self-attention focus. At the same time, several studies highlighted that ViT is computationally expensive and needs large amounts of training data. The design of the Distilled Swin-Transformer (DST) [7] empowered implementation in resource-scarce environments without much loss of performance, which was facilitated by knowledge distillation.

TransLandSeg is a novel transfer learning model designed for landslide semantic segmentation, leveraging the Segment Anything Model (SAM) through adaptive transfer learning (ATL) to enhance its performance[9]. It significantly outperforms traditional models like UNet and PSPNet on the Landslide4Sense and Bijie datasets, achieving a mean Intersection over Union (MIoU) of 88.10 percentage and an overall accuracy of 97.60 percentage on Bijie, while also excelling in detecting both small and large landslides. The research highlights the effectiveness of integrating global and local information for accurate segmentation and sets the stage for future applications of foundation models in remote sensing.

The study presents APSAM, a novel method for weakly supervised landslide extraction that utilizes the Segment Anything Model (SAM) through adaptive prompt generation (APG)[10]. By creating box and point prompts from class activation maps

(CAMs), APSAM enhances segmentation accuracy, achieving at least 3.0 percentage improvement in F1 score and 3.69 percentage in IoU over existing state-of-the-art methods on high-resolution datasets from Hong Kong and Turkey. The results indicate that APSAM effectively captures complex landslide shapes and provides precise boundaries, demonstrating its potential for improving hazard assessment in remote sensing imagery.

RSAM-Seg is a modified Segment Anything Model (SAM) designed for remote sensing image segmentation, integrating Adapter-Scale and Adapter-Feature modules to enhance segmentation capabilities without the need for manual prompts[11]. It demonstrates improved performance over the original SAM and U-Net in tasks such as cloud detection, building detection, field monitoring, and road mapping, particularly excelling in few-shot learning scenarios and identifying absent areas in datasets. Overall, RSAM-Seg achieves significant accuracy improvements across various remote sensing tasks, positioning itself as a valuable auxiliary annotation tool.

StrideNET, developed by Maitreya Shelare and colleagues, is a dual-branch transformer model designed for terrain classification and dynamic roughness extraction, presented at the 3rd IEEE International Conference on Computer Vision and Machine Intelligence in October 2024[12]. The model leverages the Swin Transformer architecture to efficiently extract features, achieving over 99 percentage accuracy on a custom dataset comprising four distinct terrain classes. Additionally, it employs statistical methods for analyzing terrain roughness, making it applicable in various fields such as environmental monitoring, land use classification, disaster response, and precision agriculture.

The study aimed to develop a hybrid CNN-transformer model, named SCDUNet++, for accurate landslide mapping using remote sensing images[6]. By employing deep transfer learning, the model was designed to enhance performance in data-poor areas, resulting in significant improvements in Intersection over Union (IoU) scores, which increased from 8.33 percentage to 27.5 percentage, and F1 scores, which rose from 6.58 percentage to 23.67 percentage after fine-tuning. The dataset utilized for this research was a multi-channel landslide dataset created from Sentinel-2 and NASADEM data, focusing on the Luding and Jiuzhaigou earthquake areas. The findings concluded that incorporating both spectral and topographic features significantly enhances the accuracy of landslide detection.

PointSAM is designed to adapt the Segment Anything Model (SAM) for remote sensing images (RSIs) by utilizing point annotations[13]. The methodology introduces Prototype-based Regularization (PBR) to address the issue of error accumulation during the segmentation process, which is particularly crucial in scenarios with dense objects. Additionally, it employs Negative Prompt Calibration (NPC) to enhance segmentation accuracy. The results demonstrate that PointSAM significantly outperforms existing methods across three different RSI datasets, namely NWPU VHR-10, HRSID, and WHU, showcasing its effectiveness in oriented object detection. The code for PointSAM is made available on GitHub, allowing for further exploration and application by the research community.

This paper introduces MergeSAM, a novel unsupervised change detection method for high-resolution remote sensing imagery, leveraging the advanced feature extraction

capabilities of the Segment Anything Model (SAM) to automatically detect complex changes such as object splitting and merging without requiring labeled training data[14] . The method was tested on a dataset of 20 pairs of Very High Resolution (VHR) images collected from 2006 to 2019 in the suburbs of Guangzhou, China, with a spatial resolution of 0.55 meters per pixel. Specific parameters for object proposal generation were set, including a point per side of 64 and an NMS threshold of 0.7, with images scaled to a maximum side length of 1600 pixels. Evaluation metrics included Precision, Recall, F1 score, Overall Accuracy (OA), and Kappa, with results showing that MergeSAM consistently outperforms existing methods, achieving a 7 percentage improvement in the F1 score compared to AnyChange while enhancing precision and maintaining high recall rates. The effectiveness of the MaskMatching and MaskSplitting strategies is emphasized, facilitating the detection of fine change parcels with fewer false detections and missed detections, thereby showcasing MergeSAM's potential to advance unsupervised change detection in remote sensing. This research was supported by the China Postdoctoral Science Foundation General Funding Project (2024M763747).

The architecture of SegFormer incorporates a multi-head self-attention module (MSA) that is crucial for capturing global dependencies among image patches or embedding vectors[15]. However, the Vision Transformer (ViT) faces significant computational challenges due to the extensive number of embedding vectors it processes. To mitigate this issue, SegFormer introduces an efficient multi-head self-attention (EMSA) mechanism that reduces the number of embedding vectors from N to N/r, where N is the product of height (H) and width (W) of the feature map, and r is a hyperparameter known as the reduction ratio. This reduction is achieved through a convolution layer, specifically using "nn.Conv2D" in PyTorch, which reshapes the feature map of size $H \times W \times C$ into a more manageable $N/r \times r \cdot C$ format, thereby streamlining the computational process while maintaining the model's effectiveness in capturing essential features for tasks such as semantic segmentation. This research focuses on developing a lightweight model for landslide detection, referred to as LDNet, which utilizes the DeepLabv3+ architecture enhanced with a dual attention mechanism. The primary goal is to address the challenges of inaccurate edge extraction in high-resolution images and the computational inefficiencies associated with traditional semantic segmentation models. The study highlights the limitations of the original DeepLabv3+ model, which employs the Xception backbone, resulting in a high number of parameters and significant resource consumption, making it less suitable for real-time detection tasks. To overcome these issues, the researchers replaced Xception with MobileNetv2, which significantly reduces the model's parameters and accelerates training speed. Additionally, the integration of the Convolutional Block Attention Module (CBAM) allows for improved feature extraction by simultaneously weighting feature maps in both channel and spatial directions, enhancing the model's ability to capture complex details in the images.

2.3 Hybrid/ Fusion Models

The paper by authors Chen et al. [16]aims to propose a new and efficient model called BisDeNet for landslide detection from optical remote sensing images. The implementation method consists of replacing the context path of the original BiSeNet with DenseNet due to its feature extraction ability and low model complexity, integrating landslide causative factors (LCFs) into the sampling dataset and using the resulting model for landslide detection because it is lighter. The framework was tested in two study areas (Zigui and Jiuzhaigou) and achieved promising results. When it comes to metrics, BisDeNet obtained high F1-scores of 0.9006 and 0.8850 in the two study areas, improving by 26.22 percentage and 1.86 percentage over the original BiSeNet. Notably, it achieves this while having the least number of parameters and the lowest memory requirements among all evaluated models. Lightweight networks tend to have lower capacity for capturing complex features and textures in highly intricate terrains, which subsequently decreases the LD accuracy and adaptability compared to non-lightweight models.

The paper authored by Ghorbanzadeh et al.[17] aims to show the potential of improving a landslide detection DL model (ResU-Net) using multi-temporal Sentinel-2 images and a DEM with an implementation of Object-Based Image Analysis (OBIA). In the first step, the ResU-Net-alone method was implemented and in the second step, a rule-based OBIA approach (land cover and image difference indices) was used to integrate the probability results from ResU-Net. This method effectively addresses the issue of fuzzy boundaries in deep learning results by utilizing a hybrid approach. The enhancement led to an over 22 percentage increase in the mean Intersection-over-Union (mIoU) of the resultant landslide map compared with the ResU-Net model. The key limitation/future research direction is that more studies need to be conducted for the seamless integration of the FCNs and OBIA method for both image and instance segmentation of landslides because only one study has been conducted.

In the study by Liu et al. [18]the remote sensing images of landslides exhibit analogous and inhomogeneous features as well as blurred boundaries, which cause low segmentation accuracy. The technique implements the AST-UNet model, an innovative deep learning architecture derived from SwinUNet structure. Two modules were added to the network: channel attention and spatial intersection (CASI) module and spatial detail enhancement (SDE) module. The CASI module is a parallel branch on the encoder that applies spatial intersection to broaden the attention range and employs a channel attention mechanism to refine attention weights. The SDE module is placed in the skip connection to enhance the decoder's focus on detailed features and improve the accuracy of landslide boundaries. The authors utilized landslide data from the Luding region in Sichuan, China, for their model experiments.The metrics for model evaluation were: F1-score, mIoU, foreground IoU (FIoU) and Hausdorff distance (HD95). The results showed that their model outperformed several state-of-the-art models like FCN, UNet and SwinUNet. The proposed AST-UNet achieved an F1-score of 90.14 percentage and mIoU of 83.45 percentage, FIoU of 70.81 percentage and an HD95 of 3.73 on the experimental datasets. The paper also noted that the model outperformed others with minimal parameters and computational complexity. The primary limitations of the study were the small size of the training dataset, which

can lead to neural network overfitting and the absence of automated raster-to-vector postprocessing techniques. Researchers identify a gap in the need to explore larger training datasets and automate post-segmentation workflows.

The method employed by Wang et al.[19] aimed to enhance the accuracy and efficiency of landslide identification by tackling challenges posed by limited samples and data complexity. Their approach entailed developing a dual-coded segmentation network (DS Net) which extracts features from both Swin Transformer and Contrastive Language-Image Pretraining (CLIP) encoders. The Swin Transformer extracts multi-scale hierarchical features and the CLIP encoder utilizes a large pre-trained model to enhance feature understanding. Two novel modules, including the prior knowledge integration (PKI) module and cross-feature aggregation (CFA) module, are designed to dynamically align local/global image features and deeply fuse domain-specific knowledge. The model was tested on a landslide identification library constructed via high-resolution remote sensing images of Beichuan Qiang Autonomous County, China and compared to SegFormer, SegNeXt, FeedFormer and U-MixFormer models. The metrics included overall accuracy (OA), precision, recall and F1-score.The results showed that DS Net performed the best with an OA of 0.926, with a precision of 0.884, a recall of 0.879 and an F1-score of 0.882. These scales were from 3.5 percentage and 7.1 percentage higher than the other models. The model has difficulty segmenting small and elongated landslides, which is likely due to the insufficient number of training samples with this morphology. Future work needs to focus on acquiring more samples of small and elongated landslides to enhance the dataset. Furthermore, an algorithm that uses morphological parameters for feature extraction should be developed to improve the detection of thin structures.

The paper by Liu et al. [20] aims to address the problem of accurately detecting landslides, specifically the visually blurred old ones, from High-Resolution Remote Sensing Images (HRSIs) where spatial features such as slopes and altitude cannot be fully extracted from 2-D images. The methodology utilizes a Siamese network without weight sharing to obtain respective texture and shape features from HRSIs and terrain features from DEMs and then fuses these features on a higher feature layer for deeper semantic feature distillation. The proposed FFS-Net model is the first multisource heterogeneous data feature-fusion-based semantic segmentation work for visually blurred landslide detection through HRSIs and DEMs. The key metrics indicated that the FFS-Net improved the segmentation accuracy of old landslides by increasing their mean intersection over union (mIoU) from 0.508 and 0.624 to 0.67. The primary limitations/research gaps that motivated the study included the considerable challenges in detecting old landslides because of their visually blurred appearance due to sediments, vegetation and human activities over time. Another one was the inability of 2-D HRSIs to sufficiently extract crucial terrain features from a visually blurred surface.

The study by Li et al.[21] aimed to tackle the limitations of single-label image classification approaches for landslide detection by introducing a novel multilabel classification (MLC) network. The methodology proposed by the authors entails a two-module network referred to as Swin-BiLSTM-FS. The Feature Representation (FR) module employs a Swin Transformer to extract the high-level semantic features from

the image and convert them into feature sequences. The second module is Category Relation Parsing (CRP) and utilizes BiLSTM to model the dependency of labels, aiming to improve the capability of detecting landslides and exploring the relationship between landslides and ground objects. For this study, the authors utilized multi-label annotation of the publicly available Bijie landslide dataset. The study employed standard category-based and example-based metrics, such as F1 score, precision and recall.The results demonstrated that the proposed model obtained the highest landslide F1 score 97.56 percentage. Additionally, the model achieved higher accuracy in both the entire MLC task and the landslide category compared to other methods that employed single-label classification or multimodal data. However, the study notes that due to the absence of adequate multilabel datasets, much effort needs to be put into creating high-quality ones for training MLC models. One of the key gaps for future studies is to extend this network to weakly supervised semantic segmentation and improve its generalizability on those specific categories. The authors conclude that their model strikes an adequate balance between performance and computational efficiency for emergency response implementation.

Authors Li et al.[22] proposed the CTHNet (CNN-Transformer Hybrid Network) so that improve landslide identification on the Loess Plateau, where complex environments cause false positives and missed detections in high-resolution remote sensing images (HRSIs). The hybrid encoder encompasses a CNN module and a multi-scale lightweight transformer (MLT) module, working in unison. The CNN module is an accelerated version of VGG-16 and designed to extract local high-dimensional features, while the MLT module captures long-range dependencies and global context. A decoder with skip connections is used to progressively restore the image details and fuse features from different scales. The authors collected a dataset of 1500 loess landslides and non-landslide samples from Liulin County, China for the experiments. The metrics used for the evaluation were precision, recall, F1-score and IoU. The results show that CTHNet obtained an F1-score of 72.81 percentage and an IoU of 57.23 percentage, which is at least 3.81 percentage higher than other CNN-based networks (U-Net, DeepLabV2, Attention-UNet, HCNet) and transformer-based networks (SETR, Swin-UNet, TransUNet, SegFormer) in F1-score and 4.56 percentage in IoU. The model was able to accurately capture both global and local landslide characteristics and suppress incorrect predictions and omissions. The main limitation of the study is that future studies should adopt InSAR data for better identification of deforming landslides. The study's contribution lies in the practical application of automatic and intelligent landslide identification, which is significant for disaster management on the Loess Plateau.

The authors Wang et al. [23] developed a model based on the Swin Transformer with multi-source data fusion to solve the challenges associated with landslide identification in environments that are more complex compared to others. The authors' methodological approach centered around implementing a landslide identification model grounded in the Swin Transformer, an advanced vision transformer that efficiently captures multi-scale features and models long-range dependencies. They developed a landslide database for Ya'an City, Sichuan Province, incorporating optical satellite images and Digital Elevation Model (DEM) data to enrich the feature set.

A novel aspect of the study was the implementation of a boundary loss function to address the class imbalance problem where landslide samples are much smaller than non-landslide samples, helping the model better learn the boundaries of the landslides. The authors compared their optimized model with three classical semantic segmentation networks, namely, PSPNet and UPerNet . The metrics derived from the Ya'an landslide database revealed that the optimized Swin Transformer model exhibited significantly enhanced performance. Specifically, it improved overall accuracy by 1.5 percentage to 2.1 percentage and F1 score by 12.4 percentage to 16.2 percentage and IoU by 14.6 percentage to 18.5 percentage over the benchmark networks. The inclusion of DEM data as a fourth channel further elevated all four main metrics across all models. Testing on the publicly available landslide dataset of Bijie City further validated the model's robustness and efficiency. The approach's major drawback is the model's high computational complexity and large size, limiting its implementation in real-time setups. A key research gap identified for future work is the exploration of lightweight and efficient model architectures.

The paper authored by Wei et al.[24] aims to develop a feature enhancement framework for highly accurate landslide detection using high-resolution images and Digital Surface Models (DSM) derived from Unmanned Aerial Vehicles (UAV). The Attention Multiscale U-Net (AMU-Net) model proposed in this study integrates an attention module to enhance landslide feature responses and a multiscale module to capture contextual information in the U-Net architecture. Moreover, the prediction employed a shifted window strategy to decrease boundary errors. The novelties of the study are the AMU-Net model itself and its performance when it comes to mean Intersection over Union (mIoU) on a benchmark dataset, which was 0. 797 on the testing dataset, which was over 2 percentage higher than other models and the highest Area Under the Curve (AUC) of 0.97. The primary limitation/research gap lies in the sampling strategy. The authors note that while the random split method yielded better performance in controlled experiments (e.g., higher mIoU of 0.792), the regional split is theoretically more appropriate for testing a model's generalization ability to new geographical regions.

The paper by authors Ghorbanzadeh et al. [25] reports the results of the 2022 Landslide4Sense (L4S) competition that supported the development of deep learning (DL) models for automatic landslide semantic segmentation on a large, globally distributed dataset of multi-source satellite images. The Kingdrone team employed a progressive label refinement-based distribution adaptation framework using Separated Normalization (SN) to address dataset distribution inconsistency. The final model ensemble (Swin Transformer, EfficientNetV2 and SegFormer encoders) achieved the winning pixel-wise F1 metric of 74 percentage . This was a significant improvement from the U-Net baseline of 59.92 percentage, which was the paper's novel contribution. They identified the following key limitations/research gaps: small objects and severe imbalance issues (landslide pixels only account for 2 percentage of the total) and distribution inconsistency among different geographical locations and time phase datasets.

The study by Rezaie et al.[26] aimed to develop a novel framework based on deep learning (DL) models for the spatial prediction of landslide-prone areas by first detecting historical landslides caused by the Taiwan Morakot typhoon in 2009 from

Sentinel-2 imagery and then generating the Landslide Susceptibility Map (LSM). The methodology comprises two stages: four models (FCNet, ICNet, RDFNet and SegNet) were employed to generate the landslide inventory map; these models are based on deep learning semantic segmentation. The second stage used two models, namely CNN-WOA and CNN-HHO, to produce the LSMs. In addition, the current study investigated the metaheuristic algorithms (namely WOA and HHO) in optimizing CNNs' parameters for better predictive accuracy of LSM. RDFNet achieved the highest F1-score (0.67), precision (0.55), recall (0.84) and accuracy (0.92) in the detection task, while CNN-HHO was the most reliable in susceptibility mapping testing with AUROC = 0.017 and RMSE = 0.132. Future studies should conduct sensitivity analysis of DL models across different study settings and incorporate other causative factors and environmental variables for better prediction.

The study by Pirasteh et al.[27] sought to improve the accuracy of landslide detection by mitigating the shortcomings of current deep learning models, which struggle to maintain equilibrium between local detail and global feature comprehension. The model employs Swin Transformer as a backbone. The researchers implemented two intricate modules, namely, multiscale feature fusion lateral connection module (MFFLCM) and local information aggregation module (LIAM). The MFFLCM integrates multiscale features from both encoder and decoder stages to bolster the model's comprehension capabilities, while the LIAM merges local information for sharpened edge delineation. The model was trained using the Binary Cross-Entropy (BCE) with Logits Loss function and tested on 1670 high-resolution images of Taiwan's Southwest region.The metrics obtained during the tests were promising, as the model exhibited a F1 score of 90.7 percentage, a high mIoU of 84.2 percentage, a recall of 91.9 percentage, a kappa of 82.6 percentage and precision of 89.9 percentage.The authors found that the model significantly reduces false positives compared to other methods. The model's high computational cost and its untested adaptability to various regions and landslide types (e.g., rainfall and earthquake-induced) were identified as the main limitations. Integrating advanced transformer architectures with customized modules to address complex geospatial challenges has immense potential for real-world landslide monitoring.

Yang et al.[28] aimed to develop a landslide detection model by integrating a transformer and convolutional neural network (CNN) to solve the problem of the small receptive field in CNNs. Their approach utilized a novel segmentation network named CTransUnet, which enhances the ResU-Net backbone with an incorporated transformer to model long-range dependencies and a Convolutional Block Attention Module (CBAM) to suppress noise. The novel contribution of this paper is that it is the first attempt to integrate a transformer into ResU-Net for the landslide detection task on a small dataset. The model was validated on two independent datasets, Bijie (rainfall-induced) and Iburi (earthquake-induced), with different geological settings.The metrics were evaluated using precision, recall, F1-score, IoU and mIoU. The model obtained an mIoU of 87.91 percentage on the Bijie dataset and an F1 score of 87.73 percentage. It achieved an mIoU of 80.23 percentage on the Iburi dataset, and an F1-score of 79.33 percentage. They concluded that their model was more accurate than the standard ResU-Net in both datasets. It pointed out two limitations, namely

that transformer-based approaches are still in their infancy for landslide detection and that the model is too complex and time-consuming for practical applications. There is a need for further research on how to effectively use and maximise the feature maps generated by CNNs for transformers to reduce the reliance on pre-trained weights for specialised images.

The vast majority of state-of-the-art solutions fall into the Hybrid/Fusion category, clearly indicating that no single model type is sufficient for landslide detection at a high level. They synergistically utilize the local feature learning ability of CNNs and the global context modeling strength of Transformers. Key synthetic innovations can be categorized into four areas: (1) Architectural blending, such as embedding a lightweight Transformer (CTHNet) [14] into a CNN encoder-decoder to jointly exploit multi-scale features; (2) Targeted feature enhancement, using specifically designed modules (LIAM/MFFLCM [19]or CASI/SDE [10]) to refine local details and preserve boundaries; (3) Data modality fusion, employing separate branches without shared weights in the backbone network (FFS-Net [12]) to extract semantic information from HRSI and DEM inputs; and (4) Knowledge integration, combining deep feature outputs with an OBIA ruleset for refinement. [9, 13]The future of the field lies in designing increasingly sophisticated architectures that utilize multiple complementary pathways to weight and fuse multimodal features flow at all levels.

MANet is a Multimodal Adapter-based Network designed for semantic segmentation in remote sensing, introducing a Multimodal Adapter (MMAdapter) to fine-tune the Segment Anything Model (SAM) for multimodal data[29]. It features a Deep Fusion Module (DFM) that integrates geographic features across scales, resulting in superior performance on the ISPRS Vaihingen and Potsdam datasets compared to existing models. This study is significant as it is the first to leverage SAM's capabilities for multimodal remote sensing tasks.

The Terrain-and-Aware Segment Anything Model (TASAM) is an advanced framework designed to enhance the capabilities of the Segment Anything Model (SAM) specifically for remote sensing image segmentation tasks[30]. The primary objective of TASAM is to address the limitations of existing models in handling complex topographical and temporal variations in satellite imagery. TASAM incorporates three innovative components: the Terrain-Aware Adapter (TA-Adapter), which integrates topographic priors from Digital Elevation Models (DEMs) into the SAM image encoder, enabling the model to perform terrain-sensitive segmentation in structurally intricate regions; the Temporal Prompt Generator (TP-Prompt), which synthesizes spatial prompts by modeling the temporal evolution of land cover from multi-date satellite observations, thereby improving the model's sensitivity to changes over time; and the Multi-Scale SAM Fusion (MS-SAM), a hierarchical token fusion strategy that enhances the segmentation of small objects and fine-grained boundaries by integrating cross-scale representations within the SAM pipeline. Extensive experiments conducted on three benchmark remote sensing datasets—LoveDA, iSAID, and WHU-CD—demonstrate that TASAM significantly outperforms both the zero-shot SAM and fine-tuned segmentation baselines, particularly in challenging environments characterized by complex terrain or dynamic temporal conditions. The results indicate that

TASAM achieves a mean Intersection over Union (mIoU) of 66.3 percentage, showcasing a substantial improvement over previous models. Additionally, the model maintains real-time inference capabilities with only a modest increase in parameters and computational load, making it a practical solution for real-world applications. The findings highlight the importance of integrating terrain and temporal context into segmentation tasks, confirming that TASAM not only enhances quantitative performance but also improves interpretability and robustness in practical scenarios. Future research directions may include the exploration of multispectral data integration to further enhance segmentation accuracy and applicability across diverse geospatial contexts.

The paper presents SCANet, a novel deep learning method aimed at enhancing landslide recognition through semantic segmentation[31]. The architecture integrates a Poolformer encoder, which is based on Transformer principles, and a Separable Channel Attention Feature Pyramid Network (SCA-FPN) decoder. The Poolformer encoder is designed to effectively capture multi-scale information from input images, improving the model's ability to extract relevant features while maintaining low computational complexity. The SCA-FPN decoder, which incorporates a separable channel attention module, enhances feature fusion and pixel prediction, making it particularly effective for semantic segmentation tasks. Extensive experiments were conducted to validate the performance of SCANet against traditional architectures, such as ResNet50 combined with various decoders like Unet, FPN, and PSPNet. The results demonstrated that SCANet significantly outperformed these existing models, achieving higher Intersection over Union (IoU) and F1-scores, indicating better segmentation accuracy. The dataset used for training and evaluation was constructed from high-resolution remote sensing images of the Loess Plateau, providing a robust basis for testing the model's effectiveness in real-world scenarios. The findings suggest that the Poolformer encoder not only improves segmentation performance but also allows for flexible integration with various decoding modules. The paper concludes that SCANet represents a significant advancement in the field of semantic segmentation for landslide detection, offering a promising approach that combines the strengths of Transformer architecture with effective feature extraction and attention mechanisms.

The research presents LandslideSegNet, a novel deep learning model designed for effective landslide segmentation using remote sensing imagery[32]. The architecture incorporates an encoder-decoder structure that utilizes Encoder-Decoder Residual (EDR) blocks, which enhance feature extraction and improve the model's ability to capture intricate details in the input data. A key component of the model is the Efficient Hybrid Attentional Atrous Convolution (EHAAC) module, which integrates asymmetric atrous convolution blocks with varying kernel sizes to extract features at multiple scales, thereby enriching the semantic information retrieved from the input feature maps. This structural arrangement not only optimizes feature recognition capabilities but also ensures that the model can focus on significant features while filtering out irrelevant information, ultimately increasing its efficiency and accuracy in detecting landslide-prone areas. The model was rigorously tested on the Landslide4Sense dataset, achieving impressive performance metrics, including 97.60 percentage accuracy and 73.65 percentage mean Intersection over Union (mIoU), demonstrating its

superiority over traditional semantic segmentation models and lightweight convolutional neural networks (CNNs). The successful application of LandslideSegNet in real-time scenarios highlights its potential in disaster management, urban planning, and traffic safety, providing valuable insights for identifying landslide-prone regions and facilitating timely decision-making processes in emergency response efforts.

The research paper titled "Conv-trans dual network for landslide detection of multi-channel optical remote sensing images" by Chen et al. (2023) presents a novel approach to landslide detection using a model called CTDNet [33]. This model integrates the strengths of ConvNeXt and Swin Transformer architectures to effectively analyze multi-channel optical remote sensing images. The authors highlight the limitations of traditional convolutional networks in capturing relationships between large objects and propose a dual-stream module, referred to as CTDBlock, which enhances feature extraction by combining shallow texture information with deep semantic information. Additionally, the paper introduces an Additional Gate Module (AGM) designed to address the loss of detail and structural information that often occurs when merging features from different network layers. The AGM selectively fuses features, improving the model's ability to detect dense and small-scale targets, which are critical in landslide detection scenarios. The experimental results demonstrate that CTDNet achieves a remarkable F1-Score of 74.71 percentage on the Landslide4Sense dataset, surpassing the performance of several existing models, including Swin-Unet and ResU-Net. The study also includes ablation experiments that analyze the impact of various modifications, such as patch size adjustments, on the model's performance. The findings indicate that the proposed CTDNet not only improves detection accuracy but also maintains efficient computational performance, making it a promising solution for real-time landslide monitoring and assessment. Overall, this research contributes significantly to the field of remote sensing and landslide detection, offering a robust framework for future studies and applications.

Recent advancements in landslide detection and prediction have leveraged deep learning methodologies, particularly through the use of modular frameworks that allow for customizable heads tailored to specific tasks, such as object detection with YOLO and classification heads for predictive analytics[34]. Notably, the Conv-LSTM model has demonstrated superior performance in seasonal landslide prediction, while Multi-Layer Perceptrons (MLPs) have excelled in forecasting peak displacements. The integration of hybrid models, particularly CNN-RNN combinations, has significantly enhanced the analysis of landslide susceptibility and mapping accuracy. The deep learning revolution has transformed the landscape of geohazard monitoring, enabling the effective processing of complex datasets and improving the capabilities of landslide detection and prediction systems. However, challenges remain, including the limited availability of training samples and the need for more comprehensive exploration of spatial dependencies, which continue to hinder the development of robust methodologies in this critical area of research. Future research directions may focus on fine-tuning large pretrained models and enhancing the integration of geophysical dynamics into predictive models to address these challenges and improve the accuracy and reliability of landslide monitoring and risk management.

This paper presents an innovative approach to detecting old landslides through a novel Iterative Classification and Semantic Segmentation Network (ICSSN) that leverages high-resolution remote sensing images (HRSIs)[35]. The primary challenge addressed is the visually blurred nature of old landslides and the limitations posed by small-sized datasets, which often lead to unsatisfactory generalization performance in traditional classification networks. To overcome these issues, the authors employ contrastive learning techniques, which have proven effective in feature extraction for both object classification and semantic segmentation tasks. The proposed ICSSN architecture consists of a feature extraction module that utilizes a weight-sharing Siamese network, allowing for simultaneous processing of two images and the concatenation of their features. This design not only enhances the network's learning capacity but also enables the generation of multiple combinations for classification predictions. The feature extraction process is further refined through the use of Atrous Spatial Pyramid Pooling (ASPP) and Squeeze-and-Excitation (SE) modules, which facilitate multi-scale feature fusion and adaptive channel weighting, respectively. The classifier module comprises a global maximum pooling layer and two fully connected layers that map the extracted features to prediction probabilities for each category. The results demonstrate a significant improvement in both object-level classification and pixel-level semantic segmentation metrics, with the F1 score increasing from 0.5054 to 0.5448 and object-level accuracy reaching 0.90. The iterative optimization method employed allows for shared feature extraction between object classification and semantic segmentation tasks, ultimately leading to a more comprehensive understanding of the landslide features and enhancing the overall detection reliability.

The research focuses on enhancing the detection of landslides through the application of deep learning techniques combined with satellite imagery, specifically utilizing data from Sentinel-1 and Sentinel-2[36]. The study introduces a heterogeneous ensemble model that significantly improves prediction performance, achieving an F1 score of 0.69 when both satellite data sources are used together. The findings indicate that employing ensemble methods not only reduces uncertainty in predictions but also enhances overall accuracy compared to single model approaches. Notably, the use of Sentinel-2 data alone results in an F1 score of 0.61, demonstrating the value of integrating multiple data sources for improved outcomes. The research emphasizes the importance of optimizing ensemble size, revealing that the best performance is achieved with an ensemble size ranging from 10 to 20 models. Beyond this threshold, the study observes a shift towards underestimation, suggesting a trade-off between the number of models and the accuracy of predictions. The results underscore the potential of using deep learning techniques for effective landslide monitoring, which can be crucial for disaster management and risk assessment in various geographic regions. The study also highlights future research directions, including the integration of additional remote sensing data sources and fine-tuning ensemble parameters to further enhance prediction models. Overall, the proposed approach offers a robust framework for automating landslide detection, contributing to the development of more effective monitoring systems.

The study proposes a landslide detection model based on the Swin Transformer, enhanced with multiscale feature fusion lateral connection and local information aggregation modules (LIAMs)

2.4 Classical Remote Sensing

The study focuses on developing a new method, referred to as WIW, aimed at effectively detecting the presence of water in wetlands by leveraging satellite data from Landsat and Sentinel[37]. The methodology involves a detailed analysis of reflectance values, employing dichotomous partitioning techniques based on ground-truth data and satellite imagery. The results indicate a high accuracy rate of 89-94 percentage in mapping water bodies, with Sentinel 2 demonstrating superior performance, achieving a kappa coefficient of 0.82. This advancement is significant as it offers a reliable tool for monitoring seasonal wetlands, which are vital for maintaining biodiversity and providing essential ecosystem services, especially in the context of ongoing climate change challenges. The findings underscore the importance of remote sensing technologies in environmental management and conservation efforts.

Table 1 Summary of CNN based models used for landslide detection

Ref No.	Core Architecture	Primary Innovation	Key Metrics	Highlighted Limitation
[1]	Mask R-CNN (ResNet-50), U-Net	Systematic comparison of instance vs. semantic segmentation models across regions via transfer learning.	Mask R-CNN superior (Dice: 0.71)	U-Net tends to "clump" landslides in proximity.
[2]	Mask R-CNN (ResNet-101)	Exploits transfer learning on small datasets of UAV-acquired digital photographs.	97% (F1-Score)	Relies heavily on high-quality annotation for small sample sizes.
[3]	Faster R-CNN (DarkNet53)	Replaced heavier backbones with DarkNet53 to stabilize training and minimize the vanishing gradient problem.	92.05% (AP), 0.80 (F1-Score)	Accuracy is lower than semantic segmentation models.
[4]	U-Net	Benchmark testing of U-Net against classic ML algorithms (SVM, RF) on small Himalayan satellite datasets.	71.12% (F1-Score)	Performance gains over ML are minimal with small datasets.
[5]	DenseNet	Optimization of DenseNet structure for efficiency and feature density when augmented with 12 environmental factors.	95.05% (F1-Score)	Demonstrated challenge of noise when merging numerous geological factors.

Table 2 Summary of Vision Transformer (ViT) based models for landslide detection

Ref No.	Core Architecture	Primary Innovation	Key Metrics	Highlighted Limitation
[6]	Swin-Transformer	Optimization of spectral input (uses RGB only) and a two-stage balanced training strategy for class imbalance.	73.99% (F1-Score)	Performance degraded when irrelevant multi-spectral bands were added.
[7]	Distilled Swin-Transformer	Implements Knowledge Distillation to create a lightweight "student" model for improved computational efficiency (lowest FLOPs).	98.16% (F1-Score)	Generalizability to other landslide types (e.g., earthquake) requires further study.

3 Background and Related Work

3.1 Remote Sensing Approaches for Landslide Detection

3.1.1 Traditional Methods (ML and OBIA)

Before deep learning became dominant, landslide detection from remote sensing (RS) imagery relied on classical machine learning (ML) and object-based image analysis (OBIA). These methods extract handcrafted features such as spectral bands, texture indices, slope, and aspect, then classify pixels or objects using ML algorithms like Random Forest, SVM, Decision Trees, or Logistic Regression. OBIA segments imagery into homogeneous objects, improving classification over pixel-based approaches. Studies show these methods achieve moderate accuracy (80–90 percentage) in relatively simple terrains and small regions. Their advantages include interpretability and lower computational requirements. However, they often struggle in complex terrains, subtle landslide boundaries, and cross-regional generalization, as performance heavily depends on feature selection and segmentation parameters. These limitations motivated the shift toward deep learning and hybrid models that automatically learn hierarchical features and capture both local and global patterns.

3.1.2 Deep Learning-Based Methods

Deep learning (DL) has transformed landslide detection by automatically learning hierarchical features from remote sensing (RS) imagery, eliminating the need for handcrafted features. Convolutional Neural Networks (CNNs) and their variants, such as U-Net, ResU-Net, and DenseNet, have been widely applied for pixel-wise landslide segmentation and boundary delineation. Studies show that CNN-based models outperform traditional ML and OBIA approaches, particularly in complex terrains and large-scale datasets. DL methods also enable integration of multi-modal data, such as optical imagery combined with Digital Elevation Models (DEM) and terrain attributes, improving accuracy and generalization. While highly effective, pure CNN models may

Table 3 Summary of Hybrid/Fusion models for landslide detection

Ref No.	Core Architecture	Primary Innovation	Key Metrics	Highlighted Limitation
[8]	BiSeNet + DenseNet	Lightweight Architecture: Replaces BiSeNet's Context Path with DenseNet blocks for parameter efficiency while retaining feature density.	90.06% (F1-Score)	Achieves high accuracy with the fewest parameters and lowest memory consumption.
[9]	ResU-Net + OBIA	Fuses CNN probability map with external Rule-Based OBIA rulesets (using indices) for post-processing refinement of landslide boundaries.	72.49% (mIoU)	Demonstrated 22% mIoU improvement over ResU-Net alone via knowledge integration.
[10]	SwinUNet	Enhanced SwinUNet with CASI (Channel/Spatial Attention) and SDE (Spatial Detail Enhancement) modules to fix boundary loss in Transformer decoder.	83.45% (mIoU)	Achieves superior boundary accuracy (lowest Hausdorff Distance) over competing U-Nets/TransUNets.
[11]	Swin + CLIP	Dual-Coded Network: Integrates "Large Model Prior Knowledge" from a pre-trained CLIP-based encoder via a PKI module for semantic enhancement.	0.882 (F1-Score)	Leverages visual-semantic understanding from general computer vision large models.
[12]	Siamese CNN (DeepLabV3+ Backbone)	Heterogeneous Data Fusion: Uses a Siamese network to separately and intelligently extract/fuse features from HRSI and DEM data for blurred old landslide detection.	0.67 (mIoU)	Designed specifically to improve detection accuracy for challenging visually blurred old landslides.
[13]	Swin-Transformer + BiLSTM	Novel application of Multi-label Classification (MLC) using BiLSTM to model inter-object dependencies (e.g., between landslide and bare soil labels).	97.56% (F1-Score)	Captures spatial relationships beyond binary classification tasks.
[14]	CNN + Transformer	CNN-Transformer Hybrid Encoder (VGG variant + MLT module) designed for joint local and global feature extraction in loess plateau regions.	72.81% (F1-Score)	Explicitly designed to handle the complex, fragmented terrain features of Loess Plateau regions.
[15]	Swin-Transformer	Fusion of multi-source data (RGB+DEM) optimized with a Boundary Loss Function (BLF) to improve boundary delineation in unbalanced data.	75.6% (F1-Score)	Highlights the importance of boundary-aware loss functions for segmentation outputs.
[16]	U-Net	Feature Enhancement Framework combining an Attention module and a Multiscale module within the convolutional U-Net structure.	0.797 (mIoU)	Demonstrates the continued value of optimising traditional CNN U-Net structures with refined attention blocks.
[17]	Swin-T, EffNetV2, SegFormer	Ensemble Strategy combining outputs of multiple leading models with advanced training techniques (separated normalization, progressive self-training, Lovasz loss).	80.41% (F1-Score)	Represents the peak performance strategy in a large-scale international benchmark competition.
[18]	CNN/Cascade/Fusion	Benchmarks different semantic segmentation frameworks (Fully Convolutional, Cascade, RDFNet) for robust landslide detection.	RDFNet: 0.67 (F1-Score)	RDFNet showed superior accuracy by fusing both RGM and Depth (DEM) features.
[19]	Swin-T + Custom Modules	Designed custom MFFLCM and LIAM modules to explicitly force multi-scale feature fusion and local information	90.7% (F1-Score)	Explicitly targets better feature fusion for small objects and fine-grained details.

struggle to capture long-range dependencies, which led to the development of hybrid CNN–Transformer architectures for enhanced global context modeling.

3.2 Transformer and Hybrid Architectures

3.2.1 Pure Transformer Models

Transformer-based architectures, such as Swin Transformer and SegFormer, have recently been applied to landslide detection for their ability to capture long-range dependencies and global contextual information. Unlike CNNs, transformers can model relationships across the entire image, improving segmentation of complex terrains and subtle landslide boundaries. Studies show that pure transformer models achieve high accuracy and better cross-regional generalization compared to traditional CNNs, particularly when combined with multi-scale feature extraction and attention mechanisms. They are especially effective for large-scale remote sensing datasets but may require substantial computational resources and large amounts of training data.

3.2.2 Hybrid CNN–Transformer Models

Hybrid CNN–Transformer models combine the local feature extraction capabilities of CNNs with the global context modeling of Transformers, making them highly effective for landslide detection. CNN layers capture fine-grained texture and edge information, while Transformer modules aggregate long-range dependencies across the image. Studies such as CTHNet and other hybrid architectures demonstrate improved segmentation accuracy, precise boundary delineation, and better cross-regional generalization compared to pure CNN or Transformer models. These models often integrate multimodal data, including optical imagery, DEM, and terrain features, and employ attention mechanisms to enhance feature representation, making them suitable for complex and heterogeneous terrains.

4 Methodology

The methodology followed in this study consists of a structured review and comparative analysis of twenty research papers related to landslide detection, monitoring, and susceptibility mapping using remote sensing, machine learning, and deep learning techniques. The goal of this methodology is to identify trends, commonly used datasets, modelling approaches, and evaluation strategies across recent studies.

First, relevant research papers were collected from reputed scientific sources including IEEE Xplore, Elsevier, Springer, MDPI, and journals focused on geoscience and remote sensing. Only papers published between 2018 and 2024 were included to ensure the review reflects current developments. Papers were selected based on their relevance to landslide mapping, segmentation, detection, or prediction using optical imagery, multispectral data, SAR, DEM, LiDAR, or multi-source datasets. Additional inclusion criteria involved the use of machine learning or deep learning models such as CNNs, U-Net, ResNet, transfer learning frameworks, or hybrid methods.

After selecting the papers, key information was extracted from each study. This included the objective of the work, dataset type and geographic region, preprocessing

steps, model architecture, training strategy, and evaluation metrics such as accuracy, IoU, F1-score, precision, recall, Kappa score, or ROC analysis. The extracted information was then categorized into common themes to identify similarities and differences across methods.

A comparative analysis was conducted to examine the performance of deep learning models versus classical machine learning approaches, understand the role of dataset quality and geographical diversity, evaluate cross-regional transferability, and study the impact of data preprocessing. The analysis also highlighted challenges reported in the selected papers, such as terrain complexity, atmospheric noise, class imbalance, and limited ground truth availability.

Finally, all extracted information was organized into summary tables for clear visual comparison. Wide-format and rotated tables were used where necessary to accommodate detailed content. This methodological framework enabled a thorough understanding of the progress, limitations, and emerging directions in landslide detection research using remote sensing and deep learning.

5 Data Sources

This study relies on multiple authoritative and high-quality data sources to ensure accuracy, reliability, and comprehensive analysis. The primary dataset consists of peer-reviewed research papers selected from reputed digital libraries such as IEEE Xplore, SpringerLink, ScienceDirect, and MDPI. These papers were chosen based on relevance to landslide detection, remote sensing, and deep learning. Additional supporting information, including satellite imagery characteristics, environmental context, and methodological descriptions, was derived from publicly available government repositories and geospatial databases. All selected sources were cross-verified to maintain consistency, and only studies with complete metadata, clear methodology, and accessible results were included for further analysis.

6 Techniques Used in Reviewed Studies

The reviewed studies utilize a diverse set of computational and analytical techniques to detect, classify, and delineate landslide regions from optical remote sensing data. These techniques can broadly be grouped into traditional machine learning, deep learning, hybrid approaches, and remote sensing preprocessing pipelines.

To begin with, traditional machine learning methods such as Support Vector Machines (SVM), Random Forest (RF), Gradient Boosting, k-Nearest Neighbors (k-NN), and Logistic Regression were applied in earlier works for pixel-based or object-based landslide classification. These models typically rely on engineered features like spectral indices, texture descriptors, elevation gradients, and slope factors extracted from satellite imagery.

In contrast, recent studies show a strong shift toward deep learning-based approaches, particularly Convolutional Neural Networks (CNN), U-Net and its variants (ResUNet, UNet++, DenseUNet), SegNet, Fully Convolutional Networks (FCN),

and encoder–decoder structures. These architectures enable end-to-end feature extraction and segmentation without manual feature engineering, making them highly suited for complex terrain analysis. Advanced deep learning enhancements such as attention mechanisms, residual learning, transfer learning, deformable convolutions, and multi-scale feature fusion were frequently used to improve landslide boundary detection and generalization across regions.

A number of studies adopted hybrid techniques, combining machine learning with deep learning—for example, using CNNs for feature extraction followed by SVM or XGBoost for final classification. Others integrated geospatial models, such as GIS-based susceptibility mapping or DEM-derived slope stability factors, into deep learning pipelines to improve accuracy in heterogeneous landscapes.

Additionally, the studies emphasized comprehensive remote sensing preprocessing techniques. Common steps included radiometric and geometric corrections, histogram equalization, noise filtering, vegetation index computation (NDVI, SAVI), shadow removal, and multi-band fusion. Some works also incorporated data augmentation, including rotation, scaling, random cropping, and illumination adjustments, to enhance model robustness.

Overall, the reviewed literature demonstrates a clear evolution from manual, feature-driven models to automated deep learning frameworks capable of capturing fine-grained spatial patterns, enabling more accurate and scalable landslide detection across diverse geographic environments.

7 Evaluation Metrics Used

The reviewed studies employ a variety of evaluation metrics to assess the performance, reliability, and robustness of landslide detection and segmentation models. Since most works focus on pixel-level or object-level classification, the commonly used metrics include Accuracy, Precision, Recall, and F1-Score, which quantify the correctness of predictions and the balance between false positives and false negatives. These metrics are essential in determining the reliability of a model in detecting landslide regions, especially in imbalanced datasets where landslide pixels are significantly fewer than non-landslide pixels.

For segmentation-based methods, more advanced metrics such as Intersection over Union (IoU) and Dice Coefficient were widely adopted to measure the overlap between predicted landslide masks and ground truth annotations. These metrics provide a fine-grained assessment of boundary accuracy and spatial consistency, which are critical for delineating landslide shapes and extents.

Several studies also used Receiver Operating Characteristic (ROC) curves, Area Under the Curve (AUC), Kappa Coefficient, and Mean Squared Error (MSE) to provide additional perspectives on model performance. These metrics help evaluate classification thresholds, model agreement beyond chance, and pixel-level prediction errors. In cross-regional experiments, some papers reported performance stability through cross-validation scores, generalization error, and domain adaptation accuracy.

Overall, the chosen evaluation metrics collectively enable a comprehensive assessment of model effectiveness, capturing both statistical accuracy and spatial precision in landslide detection tasks.

8 Discussion

The overall findings from the reviewed studies highlight significant advancements in the field of landslide detection using optical remote sensing and machine learning–based approaches. Traditional machine learning models demonstrated reasonable performance when high-quality handcrafted features were available; however, their dependency on manual feature engineering and limited generalization capacity restricted their applicability in diverse and complex terrains.

Deep learning techniques, particularly CNN and U-Net–based architectures, consistently outperformed conventional models due to their ability to automatically extract hierarchical spatial features from imagery. The incorporation of transfer learning, multi-scale feature extraction, and attention mechanisms further enhanced segmentation accuracy, particularly in regions with heterogeneous landscapes or limited training samples. Nevertheless, deep learning methods require extensive annotated datasets, which remain scarce in the landslide domain. This limitation has led to noticeable performance variations across different regions and image resolutions.

Another important observation is the heavy reliance on preprocessing techniques such as radiometric correction, vegetation index computation, data augmentation, and noise removal. These steps were found to significantly influence model performance, indicating that high-quality input data is equally crucial as the model architecture itself. In many studies, inadequate preprocessing resulted in poor boundary detection and misclassification of shadowed or vegetated regions.

While several papers reported impressive accuracy scores, the lack of standardized datasets and evaluation protocols made direct comparison challenging. Cross-regional validation was rarely performed, although the studies that attempted it revealed that models trained on one region often struggle when applied to another due to domain shifts. This gap highlights a clear need for robust domain adaptation, larger multi-regional datasets, and more generalized architectures.

Overall, the discussion underscores a clear research trend toward deeper, more sophisticated models, but also emphasizes the importance of dataset quality, generalizability, and standardized evaluation. Future research should focus on developing unified benchmarks, improving domain adaptation techniques, and integrating multi-sensor data to enhance robustness in varied environmental conditions.

9 Conclusion

This review highlights the rapid evolution of landslide detection and mapping techniques, particularly with the increasing adoption of deep learning and advanced optical remote sensing methods. Across the analyzed studies, deep neural networks—especially CNN, U-Net, and transformer-based models—demonstrated superior capability in capturing spatial patterns, recognizing complex terrain features, and improving segmentation accuracy compared to traditional machine learning approaches.

However, the findings also reveal persistent challenges such as limited annotated datasets, regional variability in landslide characteristics, and the lack of standardized

evaluation benchmarks. Many models showed strong performance within the same geographic region but struggled to generalize across diverse environments, pointing to the need for improved domain adaptation, transfer learning strategies, and comprehensive multi-regional datasets.

Despite these limitations, the overall advancements indicate a promising direction for future research. Integrating multi-sensor data, developing generalized frameworks that perform reliably across regions, and creating unified datasets for benchmarking can substantially enhance the reliability of automated landslide mapping systems. The insights from this study reinforce that while current models have achieved substantial progress, there remains significant scope for innovation to ensure robust, scalable, and globally applicable solutions for landslide hazard assessment.

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